Cascaded Convolutional Neural Network for Solar Panel Defect Classification

21CSC402P-PROJECT

Submitted by

Divyanshi Jain [RA2311026010639]
Syed Mohammed Ghouse S [RA2311026010643]
Purvee Singh [RA2311026010645]

Under the Guidance of

Dr. Illakiya T
(Assistant Professor, Department of Computational Intelligence)

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in
COMPUTER SCIENCE ENGINEERING
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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
KATTANKULATHUR- 603 203
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Department of Computational Intelligence

SRM Institute of Science & Technology

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Student Name	**: Divyanshi Jain, Syed Mohammed Ghouse S, Purvee Singh**
Registration Number	**: RA2311026010639, RA2311026010643, RA2311026010645**
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Purvee Singh, RA2311026010645

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SIGNATURE

DR. ILLAKIYA T
SUPERVISOR
ASSISTANT PROFESSOR
DEPARTMENT OF
COMPUTATIONAL INTELLIGENCE

SIGNATURE

DR. R. ANNIE UTHRA
PROFESSOR &HEAD
DEPARTMENT OF
COMPUTATIONAL INTELLIGENCE

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Divyanshi Jain
Syed Mohammed Ghouse S
Purvee Singh

ABSTRACT

Automated inspection of large photovoltaic (PV) installations is essential for sustained power output and safety, but manual thermal surveys do not scale and flat single-stage classifiers struggle with the severe class imbalance and look-alike faults of the InfraredSolarModules benchmark. This project presents Cascaded-SolarNet, a three-stage cascade of five small convolutional neural networks that classifies the 12 classes of the benchmark from single-modality infrared images of only 24 x 40 pixels. Stage 1 separates healthy from defective panels and lets healthy panels exit early; Stage 2 assigns a defect to a fault family; and Stage 3 applies one specialist network per family to name the exact fault. The system was implemented in PyTorch and evaluated on a fixed stratified held-out split of 3,000 images that is never used for training, early stopping or model selection. Removing brightness, contrast and noise augmentation, which destroys the temperature signal of thermal images, raised the from-scratch cascade from 61.7% to 79.8% accuracy, and fine-tuning an ImageNet-pretrained EfficientNet-B0 at every stage reached 83.2% accuracy, macro-F1 0.700 and 95.8% on the healthy-versus-defect decision. A flat classifier with the same backbone is essentially tied (82.6%), so the cascade did not show a clear accuracy advantage, and the family-routing stage remains the largest source of errors. Published 12-class results of up to 93.9% rely on cross-validation and pretrained features and are not directly comparable with a fixed hold-out; the report presents this comparison and its limits. Code, trained models, architecture diagrams and interactive demonstrations are provided.

Index Terms: Solar panel defect classification, infrared thermography, cascaded convolutional neural network, early exit, hierarchical classification, transfer learning, class imbalance.

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ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
BN	Batch Normalisation
CNN	Convolutional Neural Network
CV	Cross-Validation
EL	Electroluminescence
F1	Harmonic mean of precision and recall
FC	Fully Connected (layer)
GAP	Global Average Pooling
GPU	Graphics Processing Unit
IR	Infrared
NCA	Neighbourhood Component Analysis
PV	Photovoltaic
ReLU	Rectified Linear Unit
RF	Random Forest
SDG	Sustainable Development Goal
SVM	Support Vector Machine
TTA	Test-Time Augmentation
UAV	Unmanned Aerial Vehicle
ViT	Vision Transformer

CHAPTER 1

INTRODUCTION

Solar farms contain thousands of photovoltaic modules that develop defects over time, including hot spots, cracks, soiling, shadowing and diode failures. Each of these reduces power output, and some, such as undetected hot spots, create a fire risk. Manual thermal inspection is slow, expensive, labour-intensive and error-prone, and it cannot scale to utility-grade installations. Deep learning applied to drone-captured infrared (IR) imagery offers an automated alternative: an unmanned aerial vehicle with a thermal camera can image thousands of modules in one flight, and a trained classifier can label each module in milliseconds.

This project builds and evaluates such a classifier on the InfraredSolarModules benchmark, a public dataset of 20,000 IR images of 24 x 40 pixels with 12 classes (one healthy class and eleven fault types). Instead of a single network that chooses directly among all twelve classes, the project proposes a cascade of small networks that answers three easier questions in turn.

1.1 Introduction to Solar Panel Defect Classification

In an infrared image brightness encodes temperature. A healthy module heats fairly uniformly during operation, whereas a defect produces a localised temperature anomaly: a hot spot, a cold region or a characteristic thermal pattern. Classifying these patterns is therefore a pattern-recognition problem on very small images, where fine texture is absent and the position, size and shape of heat concentrations carry the information. Typical defect families are electrical faults (cells and bypass diodes), environmental conditions (soiling, shadows, vegetation) and structural damage (cracks, offline modules).

The task is difficult for three reasons. The classes are strongly imbalanced, because half of the images show healthy modules. Several fault types look alike at low resolution. And the system has to be cheap enough to run on drone hardware. These properties motivate a design that removes the dominant class first and then specialises.

1.2 Problem Statement and Description

Most existing CNN approaches use a flat, single-stage classifier that must choose directly among all classes. This design has two weaknesses. First, visually similar fault types produce overlapping features, so the network confuses them. Second, healthy panels vastly outnumber faulty ones, which biases the network towards the majority class and hurts the rare faults. The

literature reviewed in this project reports that accuracy falls from about 73% for binary tasks to about 68% for multi-class tasks across eight architectures, and that single-stage results on this benchmark stop at about 93%.

The problem addressed here is to classify a single-modality 24 x 40 pixel IR image of a PV module into one of 12 classes accurately, with a model small enough for edge use, and to understand where the errors of the proposed design come from. The literature survey identified six gaps:

- **G1.** Dual-sensor approaches are impractical. Models reaching 98% to 99% accuracy need paired thermal and visual data that a single-camera drone cannot supply.
- **G2.** Scaling the model gives negligible gains. Heavier backbones add little accuracy over lightweight ones.
- **G3.** Bounding-box detectors split thin defects. Detection models mislocalise fine crack boundaries.
- **G4.** Naive cascades add latency. Detection-plus-classification pipelines are too slow for real-time use.
- **G5.** Accuracy drops as the number of classes grows, from roughly 73% for binary to 68% for multi-class classification.
- **G6.** Similar defect types are systematically confused, and the confusion persists in flat classifiers.

1.3 Motivation

The practical cost of misclassification is concrete. A missed hot spot can become a fire; a false alarm sends a maintenance crew to a healthy module. Because most modules are healthy, a classifier that wastes effort on them is both slow and biased. A hierarchical design addresses this in a natural way: a small first network handles the easy and most frequent decision and passes only suspicious modules on, and later networks can then spend their capacity on the genuinely hard distinctions.

A second motivation is scientific. Cascades are often assumed to beat flat classifiers, but the literature offers few controlled comparisons on a small thermal benchmark. This project measures that claim with identical building blocks, identical data splits and a stage-by-stage attribution of errors, and reports the result honestly, including where the cascade does not help.

1.4 Sustainable Development Goal of the Project

The project contributes most directly to SDG 7, Affordable and Clean Energy. Solar installations that are inspected faster and more cheaply lose less energy to undetected defects, which improves the return on clean-energy investment. It also supports SDG 13, Climate Action, because every percentage point of recovered solar output displaces fossil generation, and SDG 9, Industry, Innovation and Infrastructure, through low-cost automated inspection that can run on small drones.

CHAPTER 2

LITERATURE SURVEY

This chapter reviews deep-learning approaches to PV defect detection and classification, with emphasis on infrared imagery and on the InfraredSolarModules benchmark, and derives the research gaps and objectives of the project. The studies were re-checked against their primary sources while preparing the final report; where a result had been attributed to the wrong paper in the first survey, it has been corrected below.

2.1 Overview of PV Defect Detection and Classification Methods

Automated analysis of PV imagery has moved through several generations. Early work extracted handcrafted thermal features and classified them with classical learners such as random forests. The second generation applied convolutional networks end to end, from small custom CNNs to ImageNet-pretrained VGG, ResNet and EfficientNet backbones. A third thread moved to object detection and segmentation (Faster R-CNN, Cascade R-CNN, VarifocalNet, YOLO) to localise defects. The most recent work explores cascades, ensembles, transfer learning with heavy augmentation and feature-selection pipelines, usually at the cost of extra sensors or computation.

2.2 Deep Learning Approaches for PV Defect Analysis

2.2.1 The InfraredSolarModules Benchmark and Its Baselines

Millendorf, Obropta and Vadhavkar of Raptor Maps released the InfraredSolarModules dataset at the ICLR 2020 Workshop on AI for Earth Sciences [1]. The dataset paper describes the data and lists benchmark classification results as a goal, but it reports no model accuracies. Baselines on the dataset come from later papers: a random forest at 84.0% [4], a custom CNN with augmentation at 92.5% for binary anomaly detection and 78.85% for eight selected classes [3], and an EfficientNet-B0 feature pipeline with an SVM at 93.93% for all 12 classes under 10-fold cross-validation [2].

2.2.2 Cascaded and Dual-Sensor Approaches

Pathak and Patil proposed a Binary Cascaded CNN that first separates healthy from faulty modules and then identifies the fault, fusing thermal images with co-registered visible-spectrum images [5]. It reports 99.8% precision and 98.5% accuracy, the highest in the survey,

but it needs paired dual-sensor data and substantial computation, which a single-camera drone does not provide. Barraz et al. describe a cascading ensemble with a voting CNN reaching 88% on InfraredSolarModules, as cited in [4].

2.2.3 Lightweight Networks

PV-faultNet benchmarked architectures from compact CNNs to Siamese and multi-spectral models and found that a carefully designed compact CNN reaches 93% accuracy at 8 ms per image, with larger backbones adding little [6]. The authors conclude that the bottleneck is the low information content of 24 x 40 pixel images and not model capacity, which supports small per-stage networks.

2.2.4 Detection and Segmentation Pipelines

Detection-based work extends VarifocalNet and compares it with Cascade R-CNN and YOLOv7 for localising defects in drone IR images [7]. Both detectors struggle with thin linear defects such as cracks, producing fragmented or offset boxes. Otamendi et al. combine Faster R-CNN with an EfficientNet classifier on electroluminescence images [8]; it is precise but needs powered panels imaged in darkness and adds seconds of latency. These results motivate whole-image classification for the low-resolution IR domain.

2.2.5 Transfer Learning, Ensembles and Feature Selection

A field-adapted ResNet-50 with 8x augmentation reaches 98.59% on field data but is heavy for edge use. CNN ensembles with Bacterial Foraging Optimisation for feature selection reach 98.75% with VGG-16 but must run five backbones. Tella et al. benchmarked eight CNN and ViT architectures on the ELPV benchmark and found an average of about 73% for binary and 68% for multi-class classification [9]. A four-class RGB study (normal, cracked, dusty, shadowed) reaches 96% to 98% but its coarse labels hide the real difficulty. Table 2.1 summarises the studies.

Table 2.1: Summary of the reviewed studies

#	Study	Approach	Key result	Limitation
1	Millendorf et al., 2020	Dataset paper (InfraredSolarModules)	No model results reported	Baselines come from later work
2	Pathak and Patil, 2025	Binary cascaded CNN with thermal-visual fusion	99.8% precision, 98.5% accuracy	Needs paired thermal and visual data
3	PV-faultNet, 2024	Compact CNN versus Siamese and multi-spectral	93% at 8 ms per image	Larger backbones gain little
4	Improved VarifocalNet, 2024	Cascade R-CNN, YOLOv7 comparison	High coarse detection confidence	Mislocalises thin cracks
5	Otamendi et al., 2021	Faster R-CNN + EfficientNet	High precision	Too slow for real-time

#	Study	Approach	Key result	Limitation
		on EL images		drones
6	Field-adapted ResNet-50	Transfer learning with 8x augmentation	98.59% on field data	High resource demand
7	Tella et al., 2025	8 CNN/ViT models on ELPV	About 68% multi-class vs 73% binary	Large binary to multi-class drop
8	CNN on RGB images	4-class normal/cracked/dusty/shadowed	96% to 98%	Coarse classes only
9	CNN + Bacterial Foraging Optimisation	Five pretrained CNNs with feature selection	VGG-16 + BFO: 98.75%	Extra tuning, five backbones
10	Duranay, 2023	EfficientNet-B0 features, NCA, SVM	93.93% on 12 classes (10-fold CV)	Cross-validation, pretrained features

2.3 Limitations of Existing Methods

Four limitations recur across the reviewed work. The highest accuracies depend on information a drone does not have (paired visible images, powered-panel electroluminescence) or on heavy computation (ensembles, large backbones). Detection pipelines add failure modes on thin defects. Results are reported under different protocols, mostly cross-validation, which makes them hard to compare, and near-duplicate crops in the dataset can inflate cross-validated scores. Finally, almost no study reports where a hierarchical model loses accuracy, so the benefit of cascading remains unmeasured.

2.4 Research Gaps and Contributions

Table 2.2 maps each gap to the contribution of this project.

Table 2.2: Research gaps and the contributions that address them

Gap	Research gap	Contribution
G1	Dual-sensor approaches are impractical	C4: single-modality design using only single-camera IR images
G2	Scaling the model gives negligible gains	C1: five small per-stage networks instead of one large model
G3	Bounding boxes fragment thin defects	C5: whole-image classification with no localisation step
G4	Naive cascades add latency	C2: early-exit binary front end for healthy panels
G5	Accuracy drops as classes grow	C1: hierarchical decomposition into 2-, 3- and 2-to-6-way decisions
G6	Similar defects are confused	C3: family grouping before fine classification, semantic and data-driven

2.5 Research Objectives

- **O1.** To design a three-stage cascade of small CNNs for 12-class IR solar module classification from a single sensor.

- **O2.** To implement the system reproducibly in PyTorch with a fixed stratified split, tests and published code.
- **O3.** To measure, with controlled ablations, which design choices (augmentation, input size, capacity, grouping, pretraining) matter.
- **O4.** To attribute the cascade's errors to the stage that caused them and to compare it with a flat classifier and with published results.

2.6 Plan of Action

The work followed six steps: (1) literature survey and gap analysis; (2) dataset preparation with a stratified 70/15/15 split; (3) implementation of the five-network cascade and a flat baseline; (4) controlled ablations of augmentation, input size, network capacity, family grouping and inference variants; (5) fine-tuning of an ImageNet-pretrained backbone at every stage; and (6) evaluation, error attribution, comparison with published work, architecture diagrams, demonstrations and this report.

CHAPTER 3

PROBLEM FORMULATION

This chapter defines the classification problem precisely, decomposes it into sub-problems, states the constraints under which the system is built and explains why a naive solution is not enough.

3.1 Formal Problem Definition

Let x be a single-channel infrared image of one PV module with 40 rows and 24 columns, x in R^(40 x 24) with values scaled to [0, 1]. Let the label set be Y = {y0, y1, ..., y11}, where y0 is No-Anomaly and y1 to y11 are the fault classes. The goal is to learn a function f that maps x to a label so that the expected accuracy and the macro-averaged F1 score over Y are maximised on unseen images, subject to the constraints of Section 3.3.

The label set carries a hierarchy: Y = {y0} U F_E U F_V U F_S, where the families F_E (Electrical), F_V (Environmental) and F_S (Structural) partition the eleven faults. This structure is exploited by the cascade: f(x) = g_fam(x)(x) if the triage function h(x) predicts a defect, and y0 otherwise.

3.2 Decomposition into Sub-Problems

3.2.1 Healthy Versus Defective (Triage)

The first sub-problem is the binary function h(x) in {healthy, defect}. It is the easiest decision in the system and the one that removes the dominant class. An error here is irreversible: a missed defect is never seen by later stages, and a false alarm only costs computation and a wrong label.

3.2.2 Fault Family

For images judged defective, the second sub-problem is the three-way function that selects a family. Faults that share a family should look similar to each other and different from other families, so this decision should be much easier than an eleven-way choice. Section 6 tests how well this holds.

3.2.3 Fine-Grained Subtype

Within a family, a specialist function chooses the exact fault: six classes for Electrical, three for Environmental and two for Structural. Each specialist is trained only on its own family and can spend all its capacity on the distinctions inside it.

3.3 Constraints and Assumptions

3.3.1 Dataset

The InfraredSolarModules dataset contains 20,000 images from real drone flights, 24 pixels wide and 40 pixels tall. The classes and their distribution in the stratified split used throughout this report are given in Table 3.1.

Table 3.1: Class distribution and stratified split (seed 42)

Class	Family	Total	Train	Validation	Test
No-Anomaly	Healthy	10000	7000	1500	1500
Cell	Electrical	1877	1314	282	281
Cell-Multi	Electrical	1288	902	193	193
Diode	Electrical	1499	1049	225	225
Diode-Multi	Electrical	175	123	26	26
Hot-Spot	Electrical	249	174	38	37
Hot-Spot-Multi	Electrical	246	172	37	37
Soiling	Environmental	204	143	30	31
Shadowing	Environmental	1056	739	158	159
Vegetation	Environmental	1639	1147	246	246
Cracking	Structural	940	658	141	141
Offline-Module	Structural	827	579	124	124
Total		**20000**	**14000**	**3000**	**3000**

3.3.2 Input Modality and Image Size

Only single-channel IR images are used, because a single-camera drone cannot supply paired visible images. The images are tiny, so networks must recognise where heat is concentrated and what shape it takes rather than rely on texture.

3.3.3 Whole-Image Classification

The system classifies the whole module image and does not localise defects. Detection pipelines were shown in the literature to fragment thin defects, and the dataset provides one label per crop and no boxes.

3.3.4 Evaluation Protocol

All results are measured on one fixed, stratified held-out test split of 3,000 images. The test images are never used for training, early stopping, normalisation statistics or model selection; all tuning uses a separate 3,000-image validation split.

3.4 Why the Problem Is Non-Trivial

3.4.1 Class Imbalance

Half of the images are healthy, and the rarest classes have 175 to 249 images in total (123 to 174 for training). A classifier that always answers No-Anomaly scores 50% accuracy and learns nothing about faults.

3.4.2 Look-Alike Faults

Several classes differ only in how many cells or diodes are affected (Cell and Cell-Multi, Diode and Diode-Multi, Hot-Spot and Hot-Spot-Multi), and at 24 x 40 pixels these differences are subtle.

3.4.3 Error Propagation

In a cascade, a wrong decision at an early stage cannot be corrected later. The overall accuracy of a defective image is roughly the product of the accuracies of the stages it passes through, so the early stages must be very accurate. Whether a cascade nevertheless beats a flat classifier is an empirical question, which this project answers by measurement.

3.5 Problem Statement Summary

Given single-modality 40 x 24 IR images of PV modules, build a small-network system that classifies them into 12 classes, evaluate it on a clean held-out split, attribute its errors to individual stages and compare it with a flat classifier and with published results.

CHAPTER 4

PROPOSED SOLUTION

This chapter describes the design decisions of Cascaded-SolarNet and relates each one to a gap identified in Chapter 2. The detailed architecture and implementation follow in Chapter 5.

4.1 Hierarchical Three-Stage Cascade

Cascaded-SolarNet replaces one large 12-way classifier with five small specialist CNNs arranged in three stages. Each network answers one small, well-defined question, and the answer of one stage decides which network runs next. A healthy panel passes through only one network; a defective panel passes through exactly three: the triage network, the family network and one specialist.

4.2 Early Exit for Healthy Panels

Stage 1 is a binary classifier. When it predicts No-Anomaly the pipeline outputs that label and stops. Because healthy panels are the large majority, this removes the dominant class from all later networks, handles class imbalance by structure instead of relying only on loss re-weighting, and in principle saves computation on most images.

4.3 Fault-Family Grouping

Stage 2 routes each defect to one of three families. The semantic grouping follows the physical origin of the faults: Electrical (Cell, Cell-Multi, Diode, Diode-Multi, Hot-Spot, Hot-Spot-Multi), Environmental (Soiling, Shadowing, Vegetation) and Structural (Cracking, Offline-Module). As an alternative, a data-driven grouping was derived by testing every partition of the eleven faults into three families of at least three classes and choosing the one that minimises the confusions of a trained flat network on the validation split: group A (Cell, Cell-Multi, Soiling, Vegetation, Cracking), group B (Diode, Diode-Multi, Offline-Module) and group C (Hot-Spot, Hot-Spot-Multi, Shadowing). This reduced cross-family confusions on the validation split from 264 to 87.

4.4 Two Backbones: Lightweight From Scratch and ImageNet-

Pretrained

Every stage network is available in two variants that share the cascade logic. The from-scratch variant uses small convolutional blocks and has no external data. The pretrained variant uses an ImageNet-pretrained EfficientNet-B0 with a new classification head and is fine-tuned on the stage's data. The comparison of the two variants quantifies what pretraining adds on this tiny thermal domain.

4.5 Single-Modality, Whole-Image Pipeline

The system takes only a single-channel IR image and outputs a label for the whole module. It needs no second sensor, no alignment step and no localisation, which removes the failure modes G1 and G3.

4.6 Summary of Contributions

Table 4.1: Contributions and the gaps they address

Contribution	Description	Gaps addressed
C1	Hierarchical three-stage cascade in place of a flat 12-class classifier	G2, G5
C2	Early-exit binary front end that removes the dominant No-Anomaly class	G4, imbalance
C3	Fault-family grouping (semantic and data-driven) before fine classification	G6
C4	Single-modality design using only single-camera IR images	G1
C5	Pure classification pipeline without localisation or bounding boxes	G3
C6	Controlled ablations and per-stage error attribution, with a leak-free evaluation protocol	Comparability

CHAPTER 5

ARCHITECTURE AND IMPLEMENTATION

5.1 Overview of the Implemented System

Figure 1 shows the implemented cascade. The system is implemented in Python 3.12 with PyTorch 2.11 and torchvision, scikit-learn for metrics, and Matplotlib for plots. Training and evaluation ran on an NVIDIA GeForce RTX 5060 Laptop GPU with 8 GB of memory. The code is organised as a Python package with separate modules for the class hierarchy, data handling, models, training and evaluation, plus configuration files, scripts and a test suite of 15 tests that run on synthetic data.

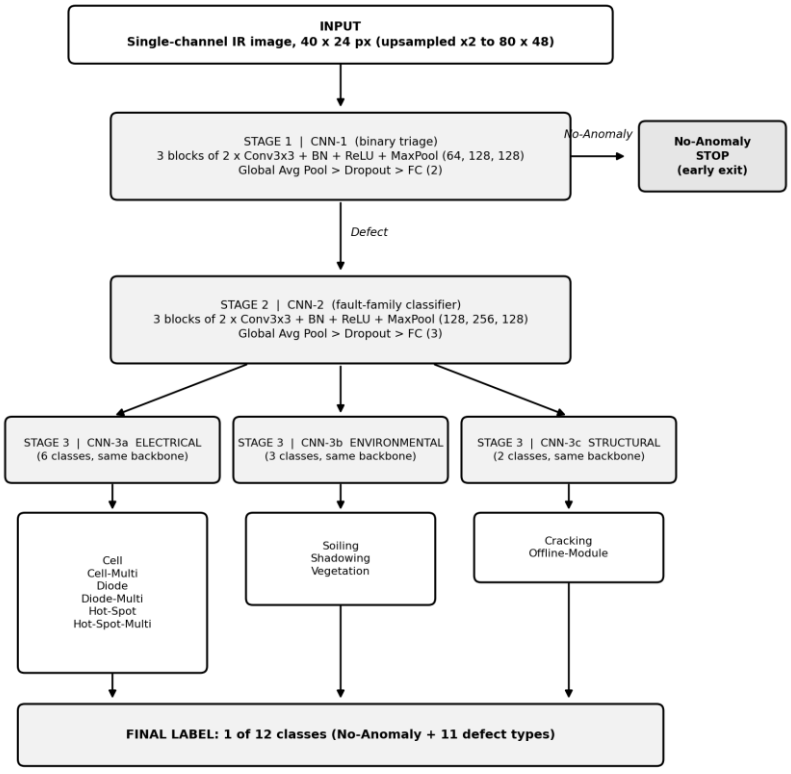


Fig. 1. The implemented three-stage Cascaded-SolarNet architecture.

5.2 Data Pipeline

5.2.1 Split and Normalisation

The 20,000 images are split once into 70% training (14,000), 15% validation (3,000) and 15% test (3,000) images, stratified by class with a fixed seed so every model sees the same split. Pixel values are scaled to [0, 1] and normalised with the mean and standard deviation of the training split only.

5.2.2 Augmentation

During training only random horizontal and vertical flips (probability 0.5 each) are applied. Brightness, contrast and noise jitter were implemented and tested but removed, because in thermal images brightness is temperature and jittering it lowered accuracy (Section 6.3). Quarter-turn rotations are not used because they would change the 40 x 24 shape.

5.2.3 Handling Class Imbalance

Each network is trained with a class-weighted cross-entropy loss whose weights are N / (K x n_r), where N is the number of training samples of that network, K the number of its classes and n_r the count of class r. Stage 1 additionally benefits from the early-exit structure itself, and the metrics reported include macro-F1, which weights every class equally.

5.3 Stage 1: CNN-1, Binary Triage

CNN-1 answers whether a module is healthy or defective. In the from-scratch variant it has three blocks of two 3 x 3 convolutions, each followed by BatchNorm and ReLU, then 2 x 2 max pooling, with 64, 128 and 128 channels, followed by global average pooling, dropout (0.2) and a fully connected layer with two outputs. The 80 x 48 upsampled input is reduced to 64 x 40 x 24, 128 x 20 x 12 and 128 x 10 x 6 feature maps. A prediction of No-Anomaly ends the pipeline (the early exit); a defect continues to Stage 2.

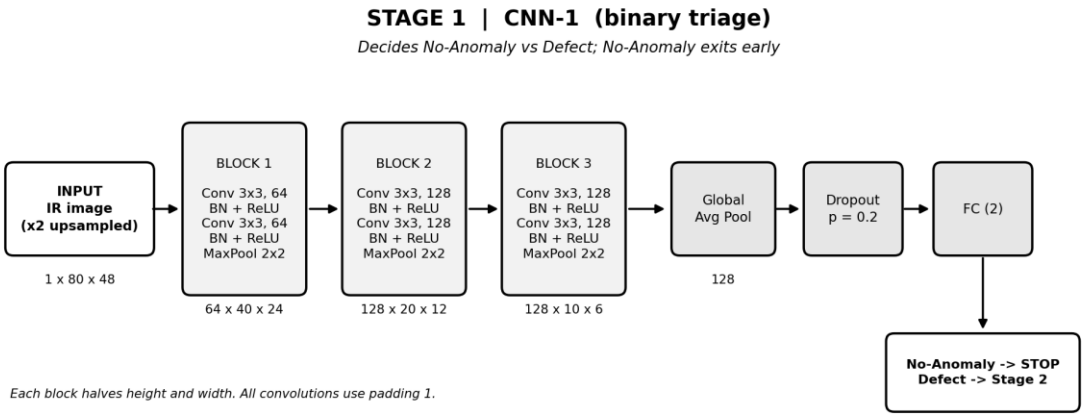


Fig. 2. Stage 1: CNN-1, binary triage (layers and feature-map sizes).

5.4 Stage 2: CNN-2, Fault-Family Classifier

CNN-2 has the same three-block design with 128, 256 and 128 channels, dropout 0.4 and three outputs. It is trained only on defective images labelled by family. Its highest-scoring output selects which specialist runs next. Faults of one family should look alike, so the decision is easier than choosing among eleven faults directly.

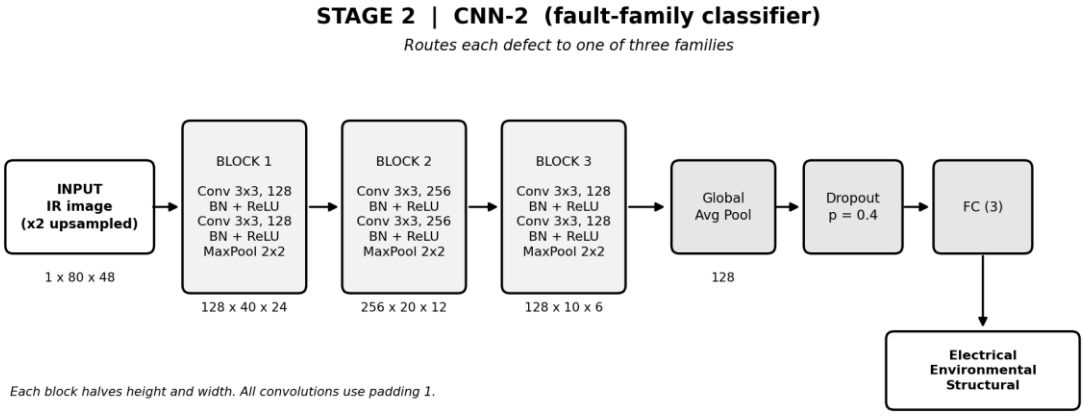


Fig. 3. Stage 2: CNN-2, fault-family classifier.

5.5 Stage 3: Fine-Grained Specialists

Three specialists share the 128-256-128 design and differ only in their output layer: CNN-3a (Electrical, six classes), CNN-3b (Environmental, three classes) and CNN-3c (Structural, two classes). Each is trained only on the defective images of its own family, so it can spend all

its capacity on the distinctions inside that family. The final label is the family's class offset plus the specialist's prediction.

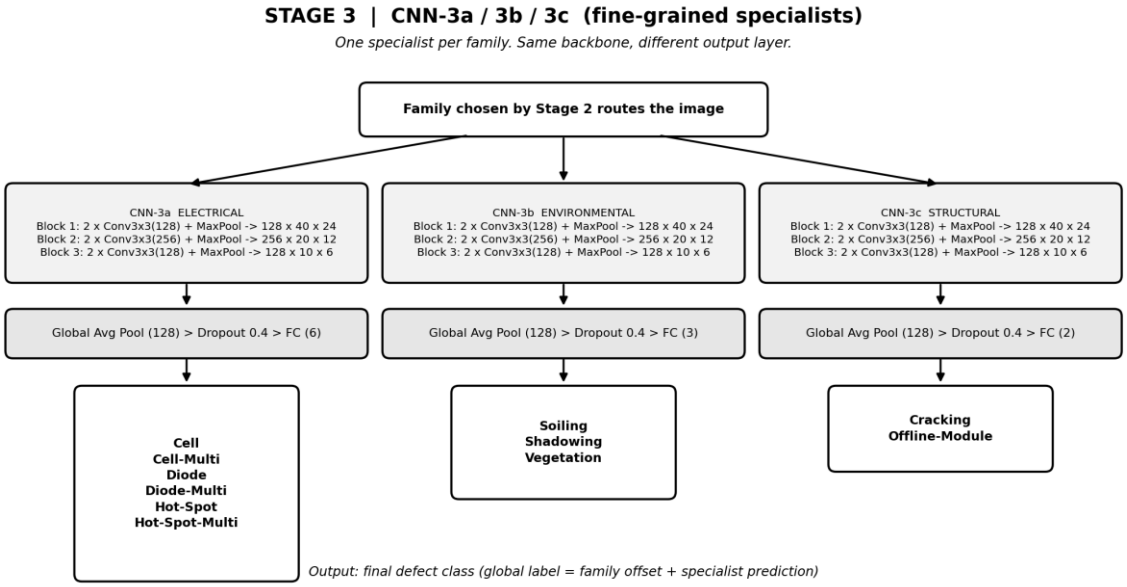


Fig. 4. Stage 3: the three specialists and their classes.

Table 5.1: Architecture summary of the five networks (from-scratch variant)

Network	Stage	Input	Layers	Output
CNN-1	1: triage	IR image, 80 x 48	3 blocks 2 x (Conv3x3, BN, ReLU) + MaxPool, 64-128-128; GAP, Dropout 0.2, FC	2 classes
CNN-2	2: family	Defective image	Same, 128-256-128; Dropout 0.4, FC	3 families
CNN-3a	3: Electrical	Routed image	Same as CNN-2, FC with 6 outputs	6 classes
CNN-3b	3: Environmental	Routed image	Same as CNN-2, FC with 3 outputs	3 classes
CNN-3c	3: Structural	Routed image	Same as CNN-2, FC with 2 outputs	2 classes

5.6 Pretrained Variant

In the pretrained variant each of the five networks is an ImageNet-pretrained EfficientNet-B0 whose classifier is replaced by dropout and a fully connected layer with the stage's number of outputs. The grayscale image is resized to 160 x 96 and repeated to three channels inside the model, so the rest of the pipeline is unchanged. The five networks hold about 20 million parameters in total, against about 6.5 million for the from-scratch cascade.

5.7 Training Procedure

Each network is trained independently on the data matching its task (ground-truth routing). Training uses the Adam optimiser with weight decay 1e-4, batch size 64, and the class-

weighted loss of Section 5.2.3. The learning rate is 5e-4 for the from-scratch networks and 3e-4 for the pretrained ones, which also use mixed-precision training. The checkpoint with the best validation macro-F1 is kept, and training stops when that metric has not improved for 25 epochs (6 for the pretrained networks). At test time the stages run in sequence on the held-out images.

5.8 Evaluation Tools and Demonstrations

The evaluation script runs the whole cascade end to end on the test split and reports accuracy, macro-F1, Cohen's kappa, per-class scores, the confusion matrix and the error attribution per stage (missed defects and false alarms at Stage 1, wrong family at Stage 2, wrong subtype at Stage 3). Further scripts reproduce the baselines, run 10-fold cross-validation, search for the data-driven family grouping and generate the figures and reports. Two demonstrations show the cascade at work: a command-line tool that draws each stage's confidence bars for chosen test images (Figure 5), and an interactive browser page in which a class can be selected and the three stages reveal their decisions one after another.

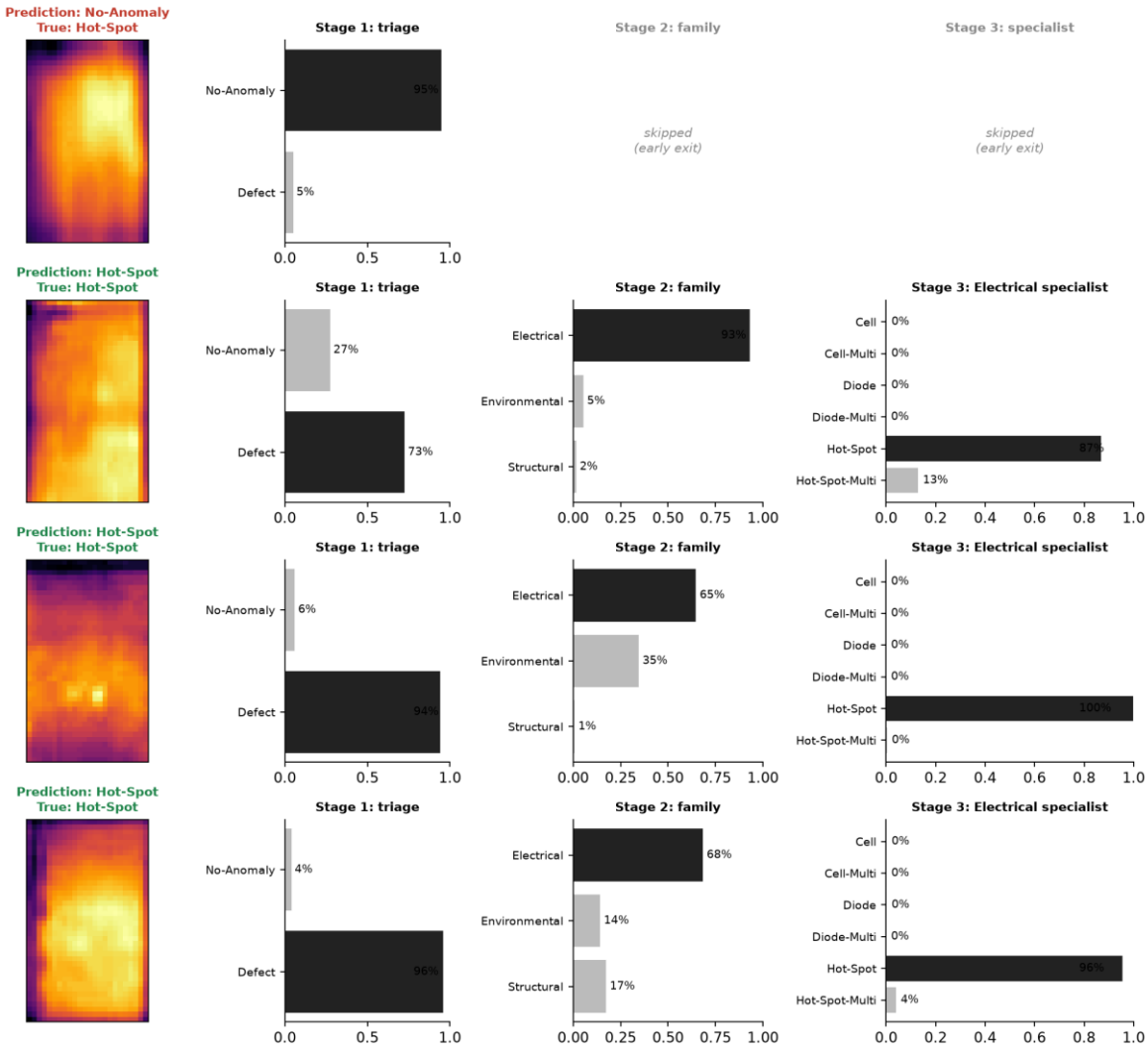


Fig. 5. Output of the live demonstration: each stage's decision and confidence for Hot-Spot test images.

5.9 Chapter Summary

The system consists of five small stage networks in two variants, a leak-free data pipeline, a class-weighted training procedure and an evaluation module that attributes every error to a stage. The next chapter evaluates it.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 Experimental Setup

6.1.1 Dataset and Protocol

All numbers are measured on the 3,000-image held-out test split of Table 3.1 and are single runs per configuration (seed 42). The test images are not used for training, early stopping, normalisation or model selection.

6.1.2 Implementation and Training Configuration

Table 6.1: Training configuration of the evaluated models

Group	Parameter	Value
Data	Split	Stratified 70/15/15 (14,000 / 3,000 / 3,000), seed 42
Data	Normalisation	Mean and standard deviation of the training split
Data	Augmentation	Random horizontal and vertical flips (p = 0.5)
From scratch	Input size	80 x 48 (2x bilinear upsampling)
From scratch	Blocks and channels	3 blocks of 2 x (Conv3x3, BN, ReLU) + MaxPool; 64-128-128 (Stage 1), 128-256-128 (others)
From scratch	Dropout	0.2 (Stage 1), 0.4 (Stage 2 and 3)
From scratch	Learning rate, patience	5e-4; 25 epochs (maximum 150)
Pretrained	Backbone	EfficientNet-B0, ImageNet weights, new head
Pretrained	Input size	160 x 96, channel repeated to 3
Pretrained	Dropout	0.2 (Stage 1), 0.3 (others)
Pretrained	Learning rate, patience	3e-4; 6 epochs (maximum 25), mixed precision
Common	Optimiser	Adam, weight decay 1e-4, batch size 64
Common	Loss	Class-weighted cross-entropy, w = N / (K x n)
Common	Model selection	Early stopping on validation macro-F1
Common	Hardware, software	NVIDIA RTX 5060 Laptop GPU; PyTorch 2.11

6.1.3 Evaluation Metrics

Accuracy is reported together with macro-F1 and Cohen's kappa, because accuracy alone hides poor performance on rare classes under class imbalance. The Stage 1 binary accuracy measures the healthy-versus-defect decision, and per-stage error counts attribute each mistake to the first stage that went wrong.

6.1.4 Configurations

Table 6.2: Definition of the evaluated configurations

ID	Variant	Modification
C1	Default	Flips plus brightness (20%), contrast (15%) and noise jitter; 40 x 24 input; 60 epochs
C2	No photometric jitter	C1 with flips only; longer schedule (100 epochs)
C3	Improved	C2 with 80 x 48 input and two convolutions per block; 150 epochs
C4	Improved2	C3 with balanced sampling and label smoothing instead of loss weights
C5	Improved3	C3 with larger Stage 1 and Stage 2 networks
C6	Data-driven families	C5 with the data-driven family grouping of Section 4.3
C7	Pretrained	EfficientNet-B0 (ImageNet) fine-tuned at every stage

6.2 Overall Performance

The best result is the pretrained cascade (C7) with 83.2% accuracy, macro-F1 0.700, Cohen's kappa 0.764 and 95.8% on the Stage 1 healthy-versus-defect decision. The best from-scratch cascade reaches 79.8% accuracy. Table 6.3 lists all configurations and baselines, and Figure 6 plots them.

Table 6.3: Test results by configuration

Configuration	Model	Accuracy	Macro-F1	Kappa	Stage 1 binary
C1 Default (jitter on)	Cascade	61.7%	0.368	0.461	81.0%
	Flat CNN	47.6%	0.379	0.357	n/a
C2 No photometric jitter	Cascade	71.1%	0.509	0.583	86.0%
	Flat CNN	68.5%	0.571	0.586	n/a
C3 Improved (80x48, deeper)	Cascade	76.0%	0.582	0.656	89.5%
	Flat CNN	76.8%	0.653	0.689	n/a
C4 Improved2 (balanced sampling)	Cascade	75.5%	0.576	0.648	89.1%
C5 Improved3 (bigger Stage 1/2)	Cascade	79.6%	0.638	0.714	93.8%
C6 Data-driven families	Cascade	79.8%	0.646	0.717	94.1%
C7 Pretrained EfficientNet-B0	**Cascade**	**83.2%**	**0.700**	**0.764**	**95.8%**
	Flat CNN	82.6%	0.708	0.761	n/a
Raw-pixel Random Forest	RF	61.6%	0.337	0.324	n/a
EfficientNet-B0 features + SVM	SVM	78.7%	0.640	0.689	90.4%

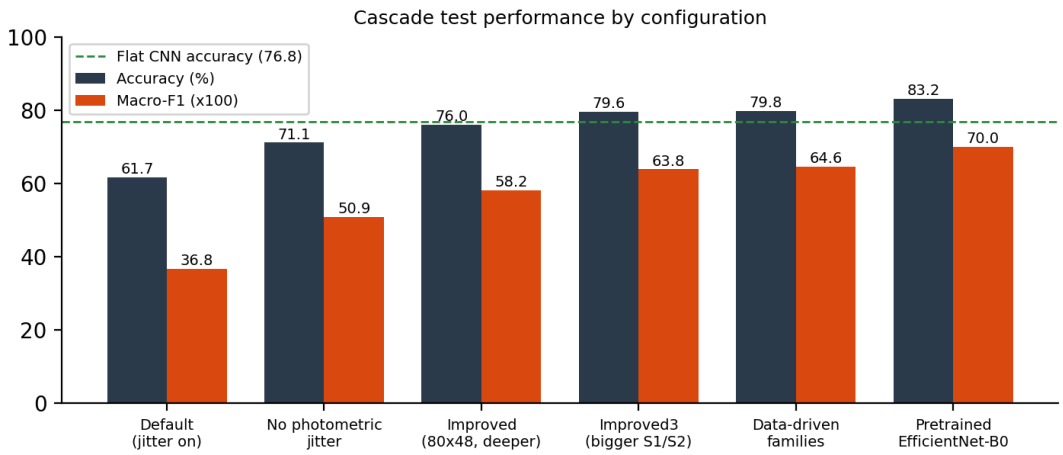


Fig. 6. Cascade test accuracy and macro-F1 by configuration (dashed line: from-scratch flat CNN).

6.3 Ablation Findings

- **Brightness and noise jitter hurt.** Removing brightness, contrast and noise augmentation raised cascade accuracy from 61.7% to 71.1% (C1 to C2). In thermal images brightness is temperature, so jittering it removes the signal that separates the classes.
- **Input size and capacity helped.** Upsampling to 80 x 48 with two convolutions per block raised accuracy to 76.0% (C3), and enlarging the Stage 1 and Stage 2 networks raised it to 79.6% (C5); Stage 1 binary accuracy rose from 89.5% to 93.8%.
- **Pretraining helped most late in the project.** Fine-tuning ImageNet-pretrained EfficientNet-B0 at every stage raised the cascade from 79.8% to 83.2% accuracy and macro-F1 from 0.646 to 0.700, and training took minutes instead of an hour.
- **The cascade and a flat classifier are essentially tied.** With the pretrained backbone the cascade scores 83.2% / 0.700 / 0.764 (accuracy, macro-F1, kappa) and the flat CNN 82.6% / 0.708 / 0.761, while the cascade has about five times the parameters. From scratch the cascade was ahead on accuracy and kappa but not on macro-F1.
- **Tried without gain.** Soft routing (multiplying stage probabilities), flip test-time augmentation, tuning the Stage 1 threshold on validation data (the optimum stayed at 0.5) and balanced sampling with label smoothing (C4).

6.4 Error Analysis by Stage

Table 6.4 attributes each test error to the first stage that went wrong.

Table 6.4: Test errors per stage (out of 3,000 images)

Configuration	Stage 1 missed defect	Stage 1 false alarm	Stage 2 wrong family	Stage 3 wrong subtype	Total
C3 Improved	233	83	294	109	719
C5 Improved3	108	79	300	126	613
C6 Data-driven	110	66	116	313	605
C7 Pretrained	**75**	**52**	**272**	**106**	**505**

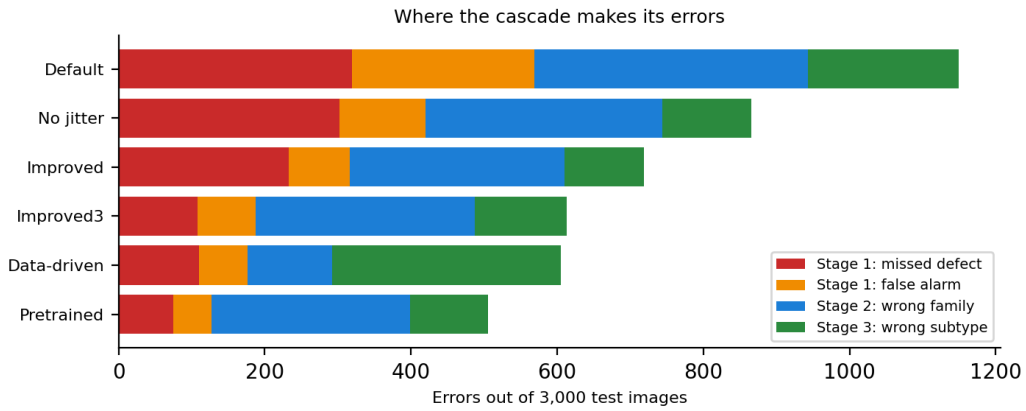


Fig. 7. Where the cascade makes its errors, by configuration.

Pretraining reduced errors at every stage compared with C5 (total 613 to 505), but Stage 2, the family decision, remains the largest source (272 of 505 errors), and an error there cannot be corrected by Stage 3. The data-driven grouping cut Stage 2 errors from 300 to 116 but moved the difficulty into Stage 3 (126 to 313 errors), so the net accuracy changed little. About 50.8% of the test images leave the cascade at Stage 1.

6.5 Per-Class Performance

The rare classes are the weak point. They have only 123 to 174 training images, and several look alike. Class weighting, stratified splitting and macro-F1 reduce but do not remove the problem.

Table 6.5: Per-class F1 on the test set

Class	Test images	Cascade, from scratch (C5)	Cascade, pretrained (C7)	Flat CNN, pretrained
No-Anomaly	1500	0.94	0.96	0.94

Class	Test images	Cascade, from scratch (C5)	Cascade, pretrained (C7)	Flat CNN, pretrained
Cell	281	0.63	0.69	0.69
Cell-Multi	193	0.47	0.52	0.49
Diode	225	0.93	0.96	0.96
Diode-Multi	26	0.82	0.87	0.91
Hot-Spot	37	0.54	0.65	0.62
Hot-Spot-Multi	37	0.57	0.65	0.62
Soiling	31	0.22	0.30	0.35
Shadowing	159	0.67	0.72	0.74
Vegetation	246	0.65	0.65	0.71
Cracking	141	0.65	0.70	0.75
Offline-Module	124	0.59	0.74	0.71

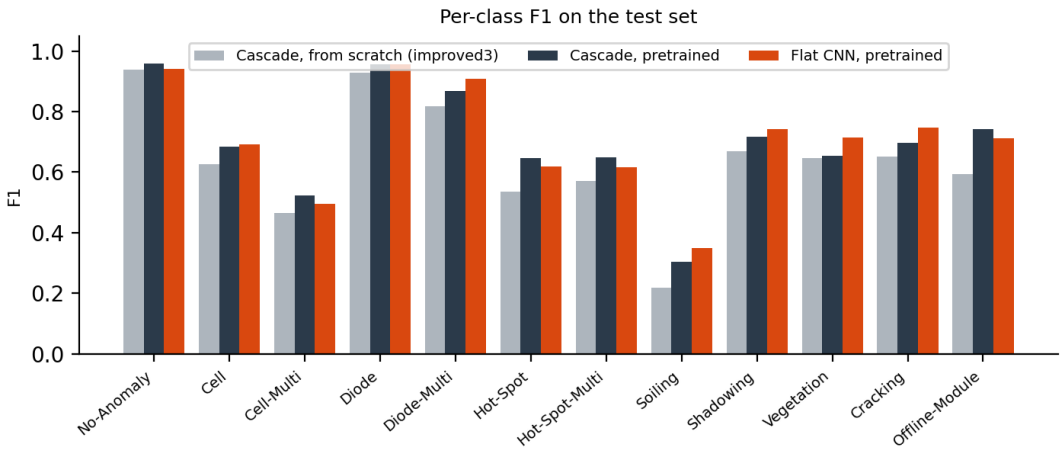


Fig. 8. Per-class F1 on the test set.

6.6 Comparison with Published Work

The dataset paper reports no model accuracies, and the figures often attributed to it (Random Forest 84%, CNN 92.5%, EfficientNet + SVM 93%) come from later papers that differ in task, protocol and features. Table 6.6 therefore places published results next to ours but does not rank them.

Table 6.6: Published results on InfraredSolarModules and results of this project

Paper	Method	Task and protocol	Reported	Checked
Duranay 2023 [2]	EfficientNet-B0 features (ImageNet), NCA selection, SVM	12 classes, 10-fold cross-validation	93.93% accuracy, 89.82% F1	Full paper read
Alves et al. 2021 [3]	Custom CNN with augmentation	Binary detection; 8 selected classes; cross-validation	92.5% binary; 78.85% for 8 classes	Abstract only
Corley et al. 2025 [4]	Random Forest	Anomaly classification, split not stated	84.0%	Paper text read
Barraz 2025; Gopalakrishnan 2024,	Cascading ensemble; NASNet + LSTM	Not stated	88%; 85%	As cited

Paper	Method	Task and protocol	Reported	Checked
as cited in [4]				
Millendorf et al. 2020 [1]	Dataset paper	-	No model results	Full paper read
Ours: pretrained cascade	**EfficientNet-B0 at each of 5 stages**	**12 classes, fixed 70/15/15 hold-out**	**83.2%, macro-F1 0.700**	**Our results**
Ours: from-scratch cascade	Five small CNNs	Same hold-out	79.8%, macro-F1 0.646	Our results
Ours: Stage 1 only (pretrained)	Binary triage	Healthy vs defect, same hold-out	95.8%	Our results
Ours: flat CNN / Random Forest	EfficientNet-B0 / raw pixels	12 classes, same hold-out	82.6% / 61.6%	Our results
Ours: Duranay-style baseline	EfficientNet-B0 features + SVM	12 classes, same hold-out	78.7%, macro-F1 0.640	Our results

- **Binary triage.** Our pretrained Stage 1 reaches 95.8% (94.1% from scratch), above the 92.5% of Alves et al. and below the 97.32% of a multi-scale transfer-learning CNN cited by Duranay. It is a single hold-out run, so we claim to be at or slightly above the 92.5% to 93% range and not a new best.
- **12 classes.** Our best, 83.2%, is below the published 12-class figures (84.0%, 85% to 88% for ensembles, 93.93%). We did not reach the 93% ceiling or the 95% target stated in the proposal. A Duranay-style pipeline run on our split scored only 78.7%, below our pretrained cascade, which suggests that the gap to 93.93% is largely due to the evaluation protocol rather than to a weaker model; this is a suggestion and not a proof.
- **Why the results are not like for like.** Duranay uses 10-fold cross-validation, a network trained end to end on the dataset and a supervised feature selection before the cross-validation, so as described, test images can influence these steps. Our test images never do. In our own split about 7% of test images have a near-identical training image (96% with the same label) and about 54% have a close one, so random folds would raise cross-validated scores. A raw-pixel Random Forest scores 61.6% on our split against 84.0% in [4], whose protocol is not stated, so part of the gap remains unexplained.

6.7 Discussion and Limitations

6.7.1 Interpretation of Findings

Three results stand out. Thermal-aware augmentation mattered more than any architectural change. Pretraining gave a large and cheap gain even on 24 x 40 pixel thermal images, which refutes the expectation in the proposal that pretrained backbones give little benefit. And the central hypothesis, that a cascade clearly beats a flat classifier, was only partly supported: the family decision is the weak link, and with equal backbones the two designs tie.

6.7.2 Limitations

- **Single runs.** Every configuration was run once with seed 42; differences below about 0.5 points are within noise and no confidence intervals were computed. A 10-fold cross-validation of the pretrained cascade was prepared but did not complete, so it is not reported.
- **Near-duplicate images.** The benchmark has no module or site identifiers, so near-identical crops can fall on both sides of any split.
- **Rare classes.** Soiling, Hot-Spot and Diode-Multi remain weak, which keeps macro-F1 well below accuracy.
- **Efficiency.** Latency was not measured reliably, because timings were taken while the GPU was busy, and the pretrained cascade has five times the parameters of the flat model, so the early exit did not give a demonstrated efficiency advantage.
- **Comparison with published work.** Protocols differ, and some sources could only be checked through abstracts or citing papers.

6.8 Chapter Summary

The best cascade reaches 83.2% accuracy with ImageNet pretraining and 79.8% from scratch, with Stage 1 at 95.8%. The cascade matches but does not clearly beat a flat classifier, the family stage is its largest error source, and published 12-class results are higher under protocols that are not comparable with a fixed hold-out.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project implemented Cascaded-SolarNet, a three-stage cascade of five CNNs for 12-class infrared solar module defect classification, with a reproducible split, tests, architecture diagrams and live demonstrations. After removing harmful photometric augmentation, enlarging the input and strengthening the front stages, accuracy rose from 61.7% to 79.8% from scratch and to 83.2% with ImageNet pretraining (macro-F1 0.37 to 0.70). Healthy-versus-defect triage reaches 95.8%, comparable to the 92.5% binary figure published for this dataset.

The claim that a cascade clearly beats a flat classifier was only partly supported. From scratch the cascade is ahead on accuracy and kappa; with the same pretrained backbone it ties a flat CNN while using five times the parameters. Error propagation is the central trade-off: a wrong decision at Stage 1 or Stage 2 cannot be undone, and the family decision remains the largest error source. Published 12-class results (84% to 93.93%) are higher than ours, but they use cross-validation and pretrained features, and the 93% ceiling and 95% target of the proposal were not reached under a stricter fixed hold-out. A practical lesson is that thermal images need thermal-aware augmentation: flips help, while brightness, contrast and noise jitter hurt.

7.2 Future Scope

- **Like-for-like evaluation.** Run several seeds and a leak-free 10-fold cross-validation, and replicate the full NCA pipeline, for a fair comparison with published results.
- **Better family routing.** Train Stage 2 jointly with the specialists or on the cascade's own routing errors, or use soft hand-offs trained end to end.
- **Rare classes.** Apply class-specific augmentation or synthetic oversampling for Soiling, Hot-Spot and Diode-Multi, and deduplicate near-identical crops before splitting.
- **Edge deployment.** Measure real latency, memory and energy on a drone-class device, and test distillation of the pretrained cascade into smaller networks.
- **Additional modalities.** Investigate RGB or contextual information from neighbouring panels, which the crop-level dataset cannot provide.

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